Students_table

Student_ID	Name	Age	Depertment
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

1. List all distinct depertments in the students table

```
SELECT DISTINCT depertment
FROM students;
```

Depertment
IT
HR
Finance

2. Get the average age of students per depertment

```
SELECT department,
AVG (age) As average_age
FROM Students
GROUP By depertment;
```

- IT : $(20+21) \div 2 = 20.5$
- HR : $(22+22) \div 2 = 22.0$
- Finence : $23 \div 1 = 23.0$

3 Show departments with more than 1 student

```
SELECT department,
COUNT(*) AS student_count
FROM Students
GROUP BY department
HAVING COUNT(*) > 1;
```

Department	Student_Count
IT	2
HR	2

4 Get all students whose age is between 21 and 23

```
SELECT *
FROM Students
WHERE age BETWEEN 21 AND 23;
```

Student_ID	Name	Age	Department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5 List all students in the IT department who are older than 21

```
SELECT *
FROM Students
WHERE Department IN ('IT', 'HR')
AND age >21
```

Student_ID	Name	Age	Department
2	Bob	22	HR
5	Eve	22	HR

Courses_table

Course_ID	Course_Name	Department	Credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6. Show total credits per department, only for departments with more than 5 total credits

```
SELECT department,
SUM(credits) As total_credits
FROM Courses
GROUP BY department
HAVING SUM (credits) > 5;
```

Department	total_Credits
IT	11

	Credits	Total
• IT	3+4+4	11
• Finance	2	2
• HR	3	3

↳ Only IT has more than 5 total credits, so total_credit will be 11.

7. List all courses that do not have 4 credits

```
SELECT *
FROM courses
WHERE credits <> 4
```

Course_ID	Course_Name	Department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

(8) Show top 3 Courses by credits in descending order

```
SELECT *
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

Coure_ID	Course_Name	Depertment	Credits
102	Python	IT	4
103	DataScience	IT	4
101	SQLBasics	IT	3

Enrollments_table

Enrollment_ID	Student_ID	Course_ID	Grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

(9) Get the maximum, minimum and average grade across all enrollments

```
SELECT *
MAX (grade) As max_grade,
MIN (grade) As min_grade,
AVG (grade) As average_grade
FROM enrollment;
```

Max_grade	Min_grade	Average_grade
90	78	84.6

(10) Count how many enrollments exist per course

```
SELECT Course_ID,
COUNT(*) As Enrollment_count
FROM enrollments
GROUP BY Course_ID;
```

Course_ID	Enrollment_Count
101	1
102	1
103	1
104	1
105	1

Salaries_table

Emplaye_ID	Name	Department	Salery	Bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

11- Find total Salary & total bonus per department

```
SELECT department,
 SUM(salary) As total_salary,
 SUM (bonus) As total-bonus
 FROM Salaries
 Group By department;
```

Department	total_salary	total_bonus
IT	122000	5000
HR	55000	4000
Finance	70000	6000

12- Show departments where average salary is above 5000

```
SELECT department,
AVG (salary) As average_salary
FROM Salaries
GROUP By department
 HAVING (salary) >55000
```

Department	Average_Salary
IT	61000
Finance	70000

13 - List employees whose salary plus bonus is greater than 60000.

```
SELECT employee_ID,
       name,
       department,
       salary,
       bonus,
  (COALESCE(salary,0) + COALESCE(bonus,0)) As total_compensation
FROM Salaries
WHERE (COALESCE(salary,0) + COALESCE(bonus,0)) > 60000
```

Employee_ID	Name	Department	Salary	Bonus	Total_compensation
1	Tom	IT.	60000	5000	65000
3	Spike	Finance	70000	6000	76000
4	Tyke	IT	62000	NULL	62000

Project_Table

Project_ID	Project_name	Department	Budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

14. Show total and average budget per department. Only include departments with average budget above 70000

```
SELECT department,
SUM(budget) As total_budget,
AVG(budget) As average_budget
FROM projects
GROUP By department
HAVING AVG (budget) > 70000;
```

Department	total_budget	average_budget
IT	270 000	135 000
Finance	80 000	80000

15. List all projects with budgets between 50000 and 120000, excluding the marketing department

```
SELECT *
FROM Projects
WHERE budget BETWEEN 50000 AND 120000
AND Department <> 'Marketing';
```

Project_ID	Project_name	Department	Budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR portal	HR	50 000